

UNIT-1 Introduction to AI

1.1. INTRODUCTION:

Definition: “Artificial Intelligence is the study of how to make computers do things, which, at the moment, people do better”. According to the father of Artificial Intelligence, John McCarthy, it is “The science and engineering of making intelligent machines, especially intelligent computer programs”. Artificial Intelligence is a way of making a computer, a computer-controlled robot, or software think intelligently, in a similar manner the intelligent humans think. Artificial Intelligence (AI) refers to the simulation of human intelligence in machines that are programmed to think and learn like humans. It involves creating intelligent systems capable of perceiving, reasoning, learning, and problem-solving. AI aims to develop computer systems that can perform tasks that typically require human intelligence, such as visual perception, speech recognition, decision-making, and natural language understanding. AI has numerous real-world applications across various domains, including healthcare, finance, transportation, cyber security, customer service, and entertainment. Its potential to automate tasks, provide data-driven insights, and enhance decision-making processes makes it a transformative technology with the ability to revolutionize industries and improve human lives. However, ethical considerations, transparency, and responsible AI development are essential to ensure AI systems are fair, unbiased, and aligned with human values.

1.2. HISTORY OF AI

- **The history of AI** can be traced back to ancient times when humans imagined and attempted to create artificial beings with human-like intelligence. However, the formal development of AI as a scientific discipline began in the mid-20th century. Here's a brief overview of the key milestones in the history of AI:
- **Dartmouth Conference (1956):** The term "Artificial Intelligence" was coined at the Dartmouth Conference, where John McCarthy and other researchers gathered to explore the possibilities of creating intelligent machines.

- **Early AI Research (1950s-1960s):** In the early years, AI researchers focused on symbolic or rule-based AI systems. Allen Newell and Herbert A. Simon developed the Logic Theorist, the first computer program capable of proving mathematical theorems. John McCarthy developed the programming language Lisp, which became a popular tool for AI research.
- **The Birth of Expert Systems (1960s-1980s):** Expert systems were developed to capture the knowledge and expertise of human experts in specific domains. These systems used rule-based reasoning and symbolic logic to mimic human decision-making processes. Examples include DENDRAL, an expert system for chemical analysis, and MYCIN, an expert system for diagnosing bacterial infections.
- **AI Winter (1970s-1980s):** Due to high expectations and the inability to deliver on them, AI faced a period of reduced interest and funding, often referred to as the "AI winter." Progress in AI research slowed, and there was general skepticism about the field's capabilities.
- **Emergence of Machine Learning (1980s-1990s):** Machine learning techniques, such as neural networks and statistical models, gained prominence during this period. The backpropagation algorithm for training neural networks was developed, and statistical approaches like decision trees and support vector machines were explored.
- **Rise of Big Data and Neural Networks (2000s-2010s):** The availability of vast amounts of data and increased computational power led to significant advancements in AI. Deep learning, a subfield of machine learning that uses neural networks with multiple layers, achieved remarkable results in areas such as image and speech recognition. Projects like IBM Watson demonstrated the potential of AI in natural language processing and question-answering.
- **Current Developments:** In recent years, AI has witnessed rapid advancements across various domains. Reinforcement learning has gained attention with breakthroughs in game playing, including Alpha Go defeating world champions in the game of Go. AI applications have become more prevalent in fields like autonomous vehicles, virtual assistants, healthcare diagnostics, and personalized recommendations. Throughout its history, AI has evolved from rule-based systems to data-driven approaches, with a shift toward more complex models and

algorithms. The field continues to evolve, with ongoing research in areas such as explainable AI, ethical considerations, and the societal impact of AI systems. It's worth noting that this is a high-level overview, and there have been numerous other significant contributions and developments in AI over the years. The field of AI is dynamic and continues to evolve, driven by advancements in technology, increasing data availability, and new algorithmic approaches.

1.3. APPLICATION OF AI:

Artificial Intelligence (AI) has a wide range of applications across various industries and sectors. Here are some notable areas where AI is being applied:

- **Healthcare:** AI is used in medical imaging for tasks such as detecting anomalies in X-rays, CT scans, and MRIs. It also enables predictive analytics for early diagnosis of diseases, personalized medicine, drug discovery, and robot-assisted surgery.
- **Finance and Banking:** AI is used for fraud detection, risk assessment, algorithmic trading, customer service chatbots, credit scoring, and financial planning. Natural language processing (NLP) helps analyze market trends and news sentiment for investment decisions.
- **Retail and E-commerce:** AI powers recommendation systems that personalize product suggestions to customers based on their preferences and browsing history. It also enables inventory management, demand forecasting, chatbots for customer support, and visual search.
- **Transportation and Autonomous Vehicles:** AI is crucial for autonomous vehicles, enabling them to perceive their surroundings, make decisions, and navigate safely. AI also optimizes logistics and route planning, traffic management, and predictive maintenance in transportation systems.
- **Manufacturing and Robotics:** AI is used for quality control, predictive maintenance, supply chain optimization, and robotic process automation. Robots and cobots (collaborative robots) perform tasks like assembly, packaging, and material handling with AI-driven intelligence.
- **Customer Service:** AI-powered chatbots and virtual assistants are used for customer support, providing 24/7 assistance, answering FAQs, and resolving common issues. Natural language understanding helps these systems interact with customers in a human-like manner.

- **Cyber security:** AI helps detect and prevent cyber threats by analyzing patterns, identifying anomalies, and predicting potential attacks. It assists in intrusion detection, malware analysis, network security, and fraud detection.
- **Energy and Utilities:** AI optimizes energy distribution, predicts electricity demand, monitors energy usage and enhances the efficiency of power grids. It enables smart grid management and facilitates renewable energy integration.
- **Education:** AI is used in adaptive learning platforms that tailor educational content and assessments to individual students' needs. It also assists in grading, plagiarism detection, and intelligent tutoring systems.

1.4. THE AI RISKS

While Artificial Intelligence (AI) offers numerous benefits and opportunities, there are also potential risks and challenges associated with its development and deployment. Here are some key AI risks to consider:

- **Bias and Discrimination:** AI systems can inadvertently perpetuate biases present in the data used to train them. If the training data contains biases based on race, gender, or other factors, the AI system may make biased decisions or predictions, leading to unfair or discriminatory outcomes.
- **Lack of Transparency and Explain ability:** Deep learning and complex AI models can be difficult to interpret and understand. Lack of transparency in AI decision-making processes raises concerns about accountability and the ability to explain how and why certain decisions are made. This is especially critical in sensitive domains like healthcare and finance.
- **Job Displacement and Economic Impact:** The automation potential of AI raises concerns about job displacement and the impact on the workforce. Certain tasks and roles may become obsolete, leading to unemployment or a shift in job requirements. It is important to consider strategies for upskilling and reskilling workers to adapt to the changing job landscape.
- **Security and Privacy Risks:** AI systems can be vulnerable to security breaches and attacks. Adversarial attacks can manipulate AI models by introducing carefully crafted inputs to deceive

or mislead the system. Additionally, the use of AI for data analysis raises concerns about privacy and the protection of personal information.

- **Ethical Considerations:** AI raises ethical dilemmas, such as the potential for AI to be used for malicious purposes or to violate privacy rights. There are ongoing discussions and debates around issues like autonomous weapons, privacy infringement, and the responsibility of AI developers and users.
- **Dependence on AI Systems:** Increasing reliance on AI systems and autonomous technologies may lead to a loss of human skills and capabilities. Dependence on AI without proper safeguards and fallback plans can create vulnerabilities and failures when AI systems encounter unforeseen situations or errors.
- **Unintended Consequences:** AI systems may exhibit behavior or make decisions that were not explicitly programmed or anticipated by their developers. Unintended consequences can arise due to biases, system errors, or complex interactions with dynamic environments, potentially leading to harmful outcomes.

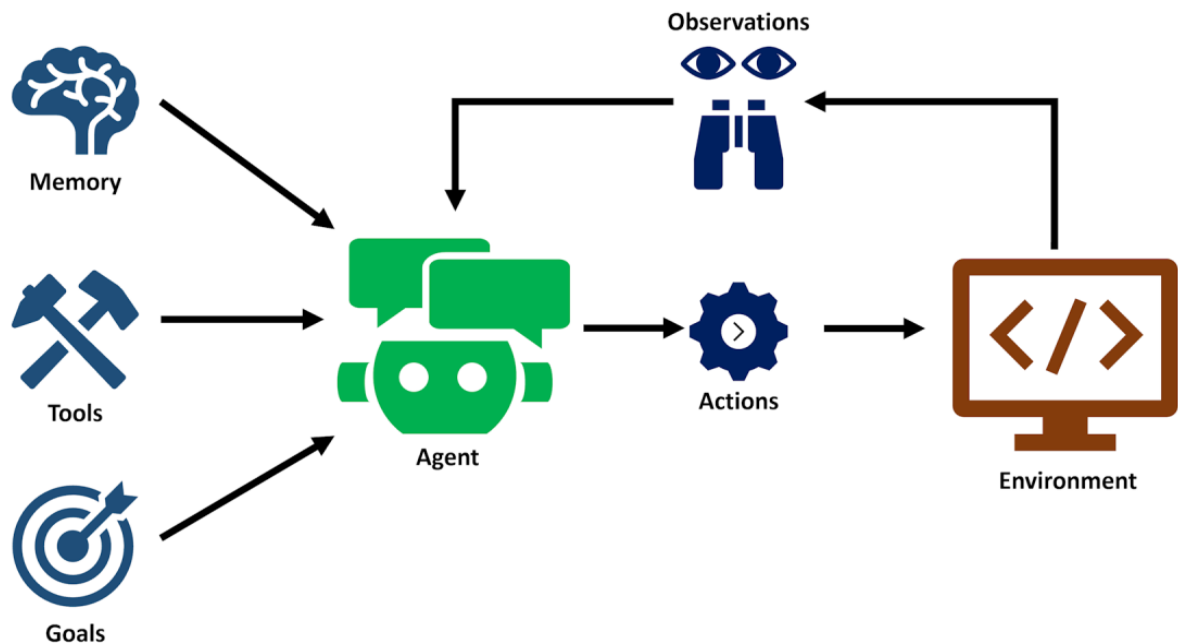
1.5. AI BENEFITS

Artificial Intelligence (AI) offers a wide range of benefits and has the potential to revolutionize various aspects of society. Here are some key benefits of AI:

- **Automation and Efficiency:** AI can automate repetitive and mundane tasks, allowing humans to focus on more complex and creative work. This leads to increased productivity, efficiency, and cost savings in industries such as manufacturing, logistics, and customer service.
- **Improved Decision-Making:** AI systems can analyze large volumes of data, identify patterns, and generate insights to support decision-making processes. This helps businesses and organizations make more informed and data-driven decisions, leading to better outcomes.
- **Enhanced Customer Experience:** AI-powered chatbots and virtual assistants enable personalized and instant customer support, improving the overall customer experience. Natural language processing capabilities allow AI systems to understand and respond to customer queries, provide recommendations, and resolve issues efficiently.

- **Advanced Data Analysis:** AI algorithms and machine learning techniques can extract valuable insights from massive datasets. This enables businesses to gain a deeper understanding of customer behavior, market trends, and operational processes. AI-powered analytics can drive innovation and competitive advantage.
- **Personalized Recommendations:** AI-based recommendation systems analyze user preferences, historical data, and behavior patterns to provide personalized recommendations. This is widely used in e-commerce, streaming services, and content platforms to enhance user experience and engagement.
- **Medical Advancements:** AI has the potential to revolutionize healthcare by aiding in early disease detection, diagnosis, and treatment planning. Machine learning algorithms can analyze medical images, genetic data, and patient records to assist doctors in making accurate diagnoses and developing personalized treatment plans.
- **Improved Safety and Security:** AI technologies are employed in surveillance systems, fraud detection, and cyber security. AI algorithms can identify anomalies, detect threats, and alert security personnel in real time, enhancing safety and reducing risks.
- **Enhanced Accessibility:** AI enables the development of assistive technologies that help individuals with disabilities. For example, AI-powered speech recognition and natural language processing allow people with mobility impairments to interact with computers and devices using voice commands.
- **Autonomous Vehicles:** AI plays a crucial role in the development of autonomous vehicles. AI algorithms process sensor data, interpret the environment, and make real-time decisions, leading to safer and more efficient transportation systems.
- **Scientific Research and Discovery:** AI aids scientific research by analyzing vast amounts of data, simulating complex systems, and identifying patterns and correlations. It accelerates the discovery process in areas such as drug development, climate modeling, and particle physics.

1.6. AGENTS AND ENVIRONMENTS



In the context of artificial intelligence (AI), agents and environments are fundamental concepts that define the interactions between an AI system and its surrounding world. Let's explore these concepts:

- **Agent:** An agent is an entity that perceives its environment, takes actions and aims to achieve specific goals. It can be a computer program, a robot, or any intelligent entity capable of sensing and acting in its environment. An agent can be simple, like a program that plays tic-tac-toe, or complex, like a self-driving car.

Types of Agent: Agents are often grouped into five classes supported by their degree of perceived intelligence and capability. of these agents can improve their performance and generate better action over time. These are given below:

- Simple Reflex Agent
- Model-Based Reflex Agent
- Goal-Based Agents
- Utility-Based Agent
- Learning Agent
- Multi-agent systems
- Hierarchical agents

- **The Functions of an Artificial Intelligence Agent**

The functions of an agent in artificial intelligence are as follows:

- To resolve complex issues using intelligent machines.
- To decide what to do in a specific situation.
- To make conclusions and take decisions.
- The perception of dynamic environmental circumstances.
- Using logic to interpret perceptions.
- To make an effort to change environmental conditions.

- **PEAS Representation**

- It may be a sort of model on which an AI agent works. once we define an AI agent or rational agent, then we will group its properties under the PEAS representation model.

It's made from four words:

- P: Performance measure
- E: Environment
- A: Actuators
- S: Sensors

Here performance measure is the objective for the success of an agent's behavior.

Agent Type	Performance Measure	Environment	Actuators	Sensors
Taxi driver	Safe, fast, legal, comfortable trip, maximize profits	Roads, other traffic, pedestrians, customers	Steering, accelerator, brake, signal, horn, display	Cameras, sonar, speedometer, GPS, odometer, accelerometer, engine sensors, keyboard

- **Properties of an agent:**

- **Percept:** The percept represents the agent's current sensory input from the environment. It could be information from sensors, such as camera images or sensor readings.
- **Action:** The action represents the agent's behavior or response to a given percept. It could be physical actions, like moving or manipulating objects, or virtual actions, like selecting a move in a game.
- **Goal:** The goal specifies what the agent wants to achieve. It could be winning a game, completing a task, or maximizing a reward.
- **Knowledge and Capabilities:** An agent may possess pre-defined knowledge, learning algorithms, or problem-solving strategies to aid in decision-making and achieving its goals.

- **Environment:** The environment is the external context in which an agent operates. It can be a physical world, a simulated environment, or a virtual domain. The environment provides the agent with sensory information and receives the agent's actions, affecting subsequent percepts. The environment can be deterministic (the next state is determined by the current state and action) or stochastic (the next state has a degree of randomness).

- **Properties of an environment:**

- **State:** The state represents the current condition or configuration of the environment, including all relevant information that determines the next percept.
- **Actions:** The environment defines the set of possible actions that an agent can take. These actions may have immediate effects or delayed consequences.
- **Transition Model:** The transition model specifies how the environment changes from one state to another based on the agent's actions.
- **Reward:** The environment provides feedback to the agent in the form of rewards or penalties, indicating the desirability of a particular state or action.
- **Interaction:** An agent interacts with the environment in a cycle of perception, action, and receiving feedback. The agent observes the current state of the environment through

percepts, selects an action based on its internal knowledge and goals, performs the action, and receives feedback in the form of new percepts and rewards. This cycle continues until the agent achieves its goal or the process is terminated.

1.7. PROBLEMS & PROBLEM SPACES

- **Problems:** A problem is defined by its initial state, goal state, and the set of actions or operators available to transition from one state to another. The initial state represents the starting point of the problem, the goal state represents the desired outcome, and the actions define the possible ways to transform the current state into a new state.
- **Problem Space:** The problem space is the set of all possible states and actions that are relevant to a given problem. It encompasses the initial state, goal state, and all the intermediate states that can be reached by applying the available actions. The problem space can be represented as a graph or a tree, where nodes represent states and edges represent actions that transition from one state to another.

1.7.1. STATE SPACE SEARCH

- A representation of the states the system can be in. For example, in a board game, the board represents the current state of the game.
- A set of operators that can change one state into another state. In a board game, the operators are the legal moves from any given state. Often the operators are represented as programs that change a state representation to represent the new state.
- An initial state.
- A set of final states; some of these may be desirable, others undesirable. This set is often represented implicitly by a program that detects terminal states.

1.7.2. THE WATER JUG PROBLEM

- The Water Jug Problem in Artificial Intelligence is a classic puzzle in AI and mathematics that focuses on optimizing the use of two or more water jugs to measure a specific quantity of water. It is a fundamental problem in the domain of optimization and decision-making. This problem

comes in various forms with different jug capacities and target measurements, making it a versatile tool for learning AI problem-solving techniques.

Classic Version

- In its classic version, the problem involves two jugs, each with a different capacity.
- The goal is to measure a specific amount of water using these jugs while adhering to certain rules and constraints.
- Let's take a simple example to illustrate the classic Water Jug Problem: You have a 3-liter jug and a 5-liter jug. The task is to measure exactly 4 liters of water.

Sample Problem Scenario

- Consider a scenario where you have a 3-liter jug and a 5-liter jug, and you need to measure precisely 4 liters of water.
- Visualize the scenario by imagining the two jugs and a water source to fill them.
- The challenge here is to determine a sequence of actions that will allow you to reach the desired measurement of 4 liters, taking into account the constraints and capacities of the jugs.

Constraints and Objectives:

- The Water Jug Problem in AI involves constraints and objectives that make it a puzzle:
Constraint 1: The jugs have limited capacities.
Constraint 2: You can only fill or pour water between the jugs or from the source.
- Objective: The goal is to measure a specific quantity of water accurately, typically by combining and transferring water between the jugs.

State Space and Action Space:

- In AI problem-solving, we work with a state space (all possible states) and an action space (all possible actions).
- In the Water Jug Problem, the state space comprises all possible configurations of water levels in the jugs.
- The action space includes the actions you can take, such as filling a jug, emptying a jug, or pouring water from one jug to another.

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Brute-Force Approach

- The brute-force approach involves systematically exploring all possible combinations of actions to solve the Water Jug Problem.
- While this method is straightforward, it may not be efficient for complex scenarios.

Simple Example and Brute-Force Solution:

- Let's consider a scenario with a 3-liter jug and a 5-liter jug, where you want to measure 4 liters of water.
- Walk participants through the brute-force solution step by step, demonstrating the actions and outcomes:

Step 1. Start with both jugs empty (0, 0).

Step 2. Fill the 3-liter jug (3, 0).

Step 3. Pour water from the 3-liter jug into the 5-liter jug (0, 3).

Step 4. Fill the 3-liter jug again (3, 3).

Step 5. Pour water from the 3-liter jug into the 5-liter jug until it's full (1, 5).

Step 6. Empty the 5-liter jug (1, 0).

Step 7. Pour the remaining water from the 3-liter jug into the 5-liter jug (0, 1).

Step 8. Fill the 3-liter jug (3, 1).

Step 9. Pour water from the 3-liter jug into the 5-liter jug until it's full (0, 4).

- This example illustrates how the brute-force approach can be used to solve the Water Jug Problem in AI by systematically testing various sequences of actions until the goal state is

reached. However, it's essential to emphasize that this method can become impractical for larger or more complex scenarios.

(i) Empty a jug



(ii) Fill a jug



(iii) Transfer



1.8. PRODUCTION SYSTEMS

- A production system is a set of rules or procedures for carrying out a task. In artificial intelligence, production systems are used to create programs that can solve problems.
- Production systems are made up of a set of production rules. Each rule has a condition and an action. The condition is tested to see if it is true. If the condition is true, the action is carried out.
- Production systems are used to create programs that can solve problems. The rules in the production system are used to find a solution to the problem. The production system can be thought of as a set of instructions for solving a problem.
- Production systems are used in many different areas of artificial intelligence. They are used in expert systems, natural language processing, and machine learning.

Components of a Production System:

- **A knowledge base:** This is a collection of facts and information that the production system can use to make decisions.
- **Inference engine:** This is the part of the system that uses the knowledge base to make decisions.

- **Working memory:** This is where the production system stores information about the current situation.
- **Control strategy:** This is the set of rules that the production system uses to decide what actions to take.

1.9. KNOWLEDGE REPRESENTATION

Knowledge representation is a crucial aspect of artificial intelligence (AI) systems as it involves the process of capturing, organizing, and modeling knowledge in a form that can be utilized by computational systems. Knowledge representation aims to enable machines to reason, understand, and make informed decisions based on the knowledge they possess. Here are some key concepts and techniques related to knowledge representation:

- **Ontologies:** Ontologies are formal representations of knowledge that define a set of concepts, their properties, and the relationships between them. They provide a structured and standardized way to organize and represent knowledge within a specific domain. Ontologies typically use a hierarchical structure and employ semantic relationships, such as subclass, part-of, and has-property, to capture the meaning and interconnections between concepts.
- **Frames:** Frames are knowledge representation structures that encapsulate information about a specific object, concept, or situation. A frame consists of a set of slots, which represent attributes or properties of the object, along with their values. Frames allow for the representation of complex and structured knowledge by capturing the properties, relationships, and behaviors associated with the object.
- **Semantic Networks:** Semantic networks represent knowledge using nodes to represent concepts or objects and edges to represent relationships between them. Semantic networks are graphical representations that can capture various types of relationships, such as inheritance, part-whole, causality, and association. They provide a visual and intuitive way to represent and reason about knowledge.
- **Rule-based Systems:** Rule-based systems represent knowledge using production rules, which consist of conditional statements (if-then rules) that define the conditions under which certain

actions should be taken. Rule-based systems are effective for representing knowledge in the form of logical and conditional relationships and are commonly used in expert systems and decision support systems.

- **Logic-based Systems:** Logic-based systems, such as predicate logic and first-order logic, represent knowledge using logical statements and rules. They provide a formal and rigorous framework for representing and reasoning about knowledge. Logic-based representations allow for the application of logical inference and deduction to conclude the given knowledge base.
- **Neural Networks:** Neural networks represent knowledge through interconnected nodes (neurons) that simulate the behavior of the human brain. Neural networks can learn from data and extract patterns and relationships, enabling them to capture implicit knowledge and make predictions or classifications based on the learned knowledge.
- **Knowledge Graphs:** Knowledge graphs represent knowledge as a graph structure, where entities are represented as nodes and relationships between entities are represented as edges. Knowledge graphs enable the representation of complex and interconnected knowledge in a flexible and scalable manner. They are often used to model large-scale knowledge bases and support sophisticated reasoning and querying capabilities.
- These are some of the common techniques and approaches used in knowledge representation in AI systems. The choice of knowledge representation technique depends on the nature of the problem, the domain being modeled, and the specific requirements of the AI application.

1.10. REASONING:

Reasoning is the mental process of deriving logical conclusions and making predictions from available knowledge, facts, and beliefs. Or we can say, "Reasoning is a way to infer facts from existing data." It is a general process of thinking rationally, to find valid conclusions. In artificial intelligence, reasoning is essential so that the machine can also think rationally as a human brain, and can perform like a human.

- **Types of Reasoning**

In AI, reasoning can be divided into the following categories:

- Deductive reasoning:
- Inductive reasoning:
- Abductive reasoning
- Common Sense Reasoning
- Monotonic Reasoning
- Non-monotonic Reasoning

- **Deductive reasoning**

The mental process of deducing logical conclusions and forming predictions from accessible knowledge, facts, and beliefs is known as reasoning. "Reasoning is a way to deduce facts from existing data," we can state. It is a general method of reasoning to arrive at valid conclusions. Artificial intelligence requires thinking for the machine to think rationally like a human brain.

Deductive reasoning is the process of deducing new information from previously known information that is logically linked. It is a type of legitimate reasoning in which the conclusion of an argument must be true if the premises are true. In AI, deductive reasoning is a sort of propositional logic that necessitates several rules and facts. It's also known as top-down reasoning, and it's the polar opposite of inductive reasoning.

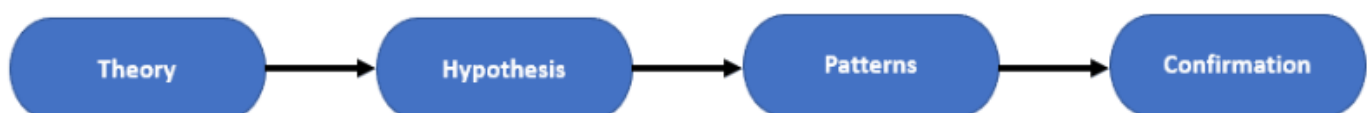
Example:

Premise-1: All the human eats veggies

Premise 2: Suresh is human.

Conclusion: Suresh eats veggies.

The general process of deductive reasoning is given below:



- **Inductive Reasoning:**

The truth of the premises ensures the truth of the conclusion in deductive reasoning.

Deductive reasoning typically begins with generic premises and ends with a specific conclusion, as shown in the example below.

Inductive reasoning is a type of reasoning that uses the process of generalization to conclude with a limited collection of information. It begins with a set of precise facts or data and ends with a broad assertion or conclusion.

Inductive reasoning, often known as cause-effect reasoning or bottom-up reasoning, is a kind of propositional logic. In inductive reasoning, we use historical evidence or a set of premises to come up with a general rule, the premises of which support the conclusion.

The truth of premises does not ensure the truth of the conclusion in inductive reasoning because premises provide likely grounds for the conclusion.

Example:

Premise: All of the pigeons we have seen in the zoo are white.

Conclusion: Therefore, we can expect all the pigeons to be white.



- **Abductive reasoning:**

Abductive reasoning is a type of logical reasoning that begins with a single or several observations and then searches for the most plausible explanation or conclusion for the observation.

The premises do not guarantee the conclusion in abductive reasoning, which is an extension of deductive reasoning.

Example:

Implication: Cricket ground is wet if it is raining

Axiom: The cricket ground is wet.

Conclusion It is raining.

- **Common Sense Reasoning**

Common sense thinking is a type of informal reasoning that can be learned through personal experience.

Common Sense thinking mimics the human ability to make educated guesses about occurrences that occur daily. It runs on heuristic knowledge and heuristic rules and depends on good judgment rather than exact reasoning.

Example:

One person can be in one place at a time.

If I put my hand in a fire, then it will burn.

The preceding two statements are instances of common sense thinking that everyone may comprehend and assume.

- **Monotonic Reasoning:**

When using monotonic reasoning, once a conclusion is reached, it will remain the same even if new information is added to the existing knowledge base. Adding knowledge to a monotonic reasoning system does not reduce the number of prepositions that can be deduced.

We can derive a valid conclusion from the relevant information alone to address monotone problems, and it will not be influenced by other factors.

Monotonic reasoning is ineffective for real-time systems because facts change in real-time, making monotonic reasoning ineffective.

In typical reasoning systems, monotonic reasoning is applied, and a logic-based system is monotonic.

Monotonic reasoning can be used to prove any theorem.

Example:

Earth revolves around the Sun.

It is a fact that cannot be changed, even if we add another sentence to our knowledge base, such as "The moon revolves around the earth" "The Earth is not round," and so on.

Advantages of Monotonic Reasoning:

In monotonic reasoning, each old proof will always be valid.

If we deduce some facts from existing facts, then it will always be valid.

Disadvantages of Monotonic Reasoning:

Monotonic reasoning cannot be used to represent real-world scenarios.

Hypothesis knowledge cannot be conveyed using monotonic reasoning, hence facts must be correct.

New knowledge from the real world cannot be added because we can only draw inferences from past proofs

- **Non-monotonic Reasoning**

Some findings in non-monotonic reasoning may be refuted if we add more information to our knowledge base. If certain conclusions can be disproved by adding new knowledge to our knowledge base, logic is said to be non-monotonic. Non-monotonic reasoning deals with models that are partial or uncertain. "Human perceptions for various things in daily life, " is a basic example of non-monotonic reasoning.

Example: Let's suppose the knowledge base contains the following knowledge:

Birds can fly

Penguins cannot fly

Pitty is a bird

In conclusion, we can say that "pitty is flying"

However, if we add another line to the knowledge base, such as "Pitty is a penguin," the conclusion that "Pitty cannot fly" is invalidated.

Advantages of Non-monotonic Reasoning:

We may utilize non-monotonic reasoning in real-world systems like Robot navigation.

We can choose probabilistic facts or make assumptions in non-monotonic reasoning.

Disadvantages of Non-monotonic Reasoning:

When using non-monotonic reasoning, old truths can be negated by adding new statements.

It can't be used to prove theorems.

1.11. AI Techniques

- AI techniques refer to the various methods and algorithms used to develop and enhance artificial intelligence systems. These techniques enable machines to perceive, reason, learn, and make decisions similar to human intelligence. Here are some commonly used AI techniques:
- **Machine Learning (ML):**
 - Machine Learning (ML) is a subfield of artificial intelligence (AI) that focuses on the development of algorithms and models that enable computers to learn and make predictions or decisions without being explicitly programmed. ML algorithms learn from data and improve their performance over time through experience.
- **Deep Learning:**
 - Deep Learning is a subset of machine learning that focuses on the development and application of artificial neural networks, particularly deep neural networks with multiple layers. Deep learning models are inspired by the structure and function of the human brain, and they are designed to learn and extract hierarchical representations of data.
- **Natural Language Processing (NLP):**
 - NLP focuses on enabling computers to understand, interpret, and generate human language. It involves techniques like sentiment analysis, language translation, text classification, and question-answering systems.
 - Natural Language Processing (NLP) is a subfield of artificial intelligence (AI) that deals with the interaction between computers and human language. It involves the development of algorithms and models that enable computers to understand, interpret, and generate natural language text or speech.
- **Computer Vision:**
 - Computer vision aims to enable machines to interpret and understand visual data from images or videos. Techniques such as object detection, image segmentation, and image recognition are used to analyze and extract information from visual content.

- Computer Vision is a field of study within artificial intelligence (AI) that focuses on enabling computers to understand and interpret visual information from digital images or videos. It involves developing algorithms and models that can analyze, process, and extract meaningful information from visual data.
- **Reinforcement Learning:**
 - Reinforcement learning involves training an agent to make sequential decisions in an environment to maximize rewards. The agent learns through trial and error, receiving feedback in the form of rewards or penalties for its actions.
 - Reinforcement Learning (RL) is a subfield of machine learning that focuses on how an agent can learn to make sequential decisions in an environment to maximize its cumulative reward. It involves learning through interaction with an environment, where the agent takes actions, receives feedback or rewards, and learns from the consequences of their actions.
- **Expert Systems:**
 - Expert systems are rule-based AI systems that use knowledge from human experts to make decisions or provide recommendations in a specific domain. They employ if-then rules and knowledge representation techniques to mimic human expertise.
 - Expert Systems, also known as knowledge-based systems, are a branch of artificial intelligence (AI) that aims to capture and utilize the knowledge and expertise of human experts in specific domains. These systems emulate the decision-making and problem-solving abilities of human experts by representing their knowledge in a computer program.